

Literature status: De Pierro’s conjecture

Source arXiv:1304.7078; answer arXiv:1801.03216

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Status: literature already answered (negative).

Original question

Baillon, Combettes, and Cominetti [1] discuss De Pierro’s conjecture in Section 3 (source PDF p. 10): as the relaxation parameter $\varepsilon \downarrow 0$, the limit cycles of under-relaxed cyclic projections onto finitely many closed convex sets should shrink to a least squares solution. They state that the conjecture remained open for general convex sets, and later prove it for $m = 2$ closed convex sets in Euclidean spaces (Corollary in Section 3.3, source PDF p. 13).

Exact later answer

Cominetti, Roshchina, and Williamson [2] explicitly identify the same De Pierro conjecture and give a counterexample. Their construction consists of three compact convex sets in $\mathbb{R}^3$ for which the ε -under-relaxed cycles do not converge as $\varepsilon \downarrow 0$. Hence the unrestricted conjecture is false.

This is an explicit same-problem answer known to the supporting authors, not an agent-derived implication. It does not alter the original paper’s positive results under additional hypotheses or its complete two-set Euclidean theorem.

Search evidence

An exact-title and keyword search for “De Pierro’s conjecture”, “under-relaxed cyclic projections”, and “counterexample” located arXiv:1801.03216. Its title and abstract explicitly state the disproof, so no new mathematical attempt was pursued.

References

- [1] J.-B. Baillon, P. L. Combettes, and R. Cominetti, *Asymptotic behavior of compositions of under-relaxed nonexpansive operators*, arXiv:1304.7078 (2013).
- [2] R. Cominetti, V. Roshchina, and A. Williamson, *A counterexample to De Pierro’s conjecture on the convergence of under-relaxed cyclic projections*, arXiv:1801.03216 (2018).